

Jireh Velasquez

jirehevelasquez@gmail.com | 805-819-1476 | Lompoc, CA

Active DoD Secret Clearance | U.S. Citizen | ITAR-Trained | Bilingual: English & Spanish

EDUCATION

California Polytechnic State University, San Luis Obispo

Exp. Grad: Jun 2026

B.S. Mechanical Engineering, Mechatronics Concentration | GPA: 3.64 |

Coursework: Mechatronics, Robotics, Propulsion Systems, Aerospace Structures, Thermo-Fluids, Composites

EXPERIENCE

Lockheed Martin Aeronautics - Skunk Works | Air Vehicle Co-op & Intern

Aug 2025 - Mar 2026

- Owned 5+ flight-critical aerospace hardware packages from concept through Phase 1/2 reviews, leading design decisions, integration readiness, and review-board defense in a classified environment.
- Authored 25+ L1/L2/L3 system requirements with subsystem traceability, verification planning, and compliance to program baselines in ITAR/export-controlled development.
- Supported integration and test-readiness reviews across 3+ concurrent aerospace projects; identified subsystem interface gaps, documented risks/discrepancies, and drove closure before milestones.
- Coordinated with design, systems, manufacturing, and program management teams to resolve technical issues, maintain configuration discipline, and prepare hardware packages for approval.

Lockheed Martin Aeronautics - Skunk Works | Chief & Systems Engineering Intern

Summer 2024

- Collaborated across Systems Engineering, Airframe, Project Management, and Chief Engineering teams on classified aerospace integration, configuration control, and test-readiness workflows.
- Created, traced, and verified system-level requirements using Cameo MBSE and GitLab; authored technical documentation and systems proposal content for cross-functional stakeholders.
- Applied ITAR training and export-control practices while handling technical data, requirements artifacts, and documentation in a controlled aerospace engineering environment.

A&P Mechanic Assistant | Automotive, Electrical & Fabrication Mechanic

2017 - 2024

- Assisted licensed A&P mechanic with aircraft airworthiness work, including Cessna 172H landing gear assembly, steering linkage installation, engine rebuild support, and inspection of flight-critical parts.
- Completed vehicle rewiring, electrical troubleshooting, soldering, Arduino-based projects, and PC board testing/assembly; developed practical embedded hardware and signal-debugging skills.
- Troubleshoot hydraulic, pneumatic, electrical, and mechanical systems across aircraft, diesel, agricultural, and field equipment platforms; resolved leaks, pressure issues, wiring faults, and readiness problems.
- Fabricated/repaired hardware using MIG/TIG/Stick welding, composites/fiberglass layups, machining support, Magnaflux NDI, and hands-on mechanical assembly practices.

PROJECTS

Autonomous Quadrotor Localization & PID Control - ArduPilot / MAVLink / Python

- Built and integrated a 4-motor quadrotor UAV from the ground up, combining flight hardware, ArduPilot configuration, telemetry communication, and autonomous control logic.
- Used MAVLink protocols to communicate with the drone through a backpack telemetry system, enabling command, telemetry, and feedback data exchange.
- Integrated an Xbox 360 depth camera with a Python tracking script to estimate the drone position in 3D space, then fed external position data back for autonomous PID correction.
- Debugged cross-domain integration issues involving motors, flight controller setup, telemetry, camera sensing, Python software, control response, and hardware/software communication.

Compressor Rotor Blade Press Fit Study - Solar Turbines

- Authored full test procedures for setup, execution, data collection, and acceptance criteria; built and validated ANSYS FEA models with manufactured test specimens and spectrum-analyzer testing.
- Conducted impact hammer testing under fixed/free boundary conditions; analyzed test data against acceptance criteria and translated results into actionable engineering recommendations.

Rocket Nozzle Design, Analysis & Manufacture - Cal Poly Space Systems Club

- Designed, manufactured, and tested propulsion hardware around flow performance, contraction/expansion ratios, thermal survivability, and test-data validation for a live-fire propulsive system.
- Conducted SRR/PDR/CDR reviews, defending propulsion design decisions, analysis results, manufacturability, and test-readiness logic in a launch-vehicle-style review process.

TECHNICAL SKILLS

Systems Integration & Test: HITL test-rig support, integration readiness, test procedures, verification planning, data collection, signal validation, discrepancy documentation, troubleshooting, design reviews, configuration-controlled engineering

UAV / Embedded / Controls: ArduPilot, MAVLink, PID control, telemetry, Python, Arduino, MATLAB, embedded systems fundamentals, computer vision tracking, closed-loop feedback, autonomous correction

Avionics / Electrical / Hardware: Wiring harness fabrication, soldering, continuity checks, PC board testing, electrical troubleshooting, communication protocol exposure, sensor integration, mechanical assembly

CAD / Analysis / Software: CATIA V5, 3DX, ANSYS, SolidWorks, Cameo MBSE, GitLab/GitHub, MATLAB, Python, Arduino

Fabrication & Lab: MIG/TIG/Stick welding, machining support, composites/fiberglass, Magnaflux NDI, impact hammer testing, spectrum analyzer testing, aircraft maintenance support, hydraulic/pneumatic troubleshooting

Documentation & Compliance: System requirements, test procedures, work instructions, SRR/PDR/CDR packages, anomaly investigation, discrepancy reporting, MBSE traceability, ITAR training

CLEARANCE & ADDITIONAL

Active DoD Secret Clearance | U.S. Citizen | ITAR-trained / export-controlled environments | FCC Amateur Radio License | PADI Open Water Diver | Tae Kwon Do Instructor, 1st & 2nd Degree Black Belt | 4-Year Varsity Soccer